Scarcity and Resource Allocation

AP Microeconomics · Unit 1 · Session 01

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Unit 1 · Session 01

A fixed resource forces a choice

A town has a fixed amount of land, labor, and time.

It chooses to produce more school meals.

What must happen to another use?

A fixed resource forces a choice

Today we will answer

How does scarcity create a choice, a trade-off, and an opportunity cost?

- identify the scarce resource;
- explain the allocation decision;
- name the best alternative given up.

Official targets for Session 01

By the end of the lesson

I can connect a limited resource to a specific choice and a specific foregone alternative.

- Define resources and explain why **scarcity creates trade-offs**.
- Identify **land, labor, and capital**; recognize that established knowledge may be non-rival.
- Answer **what, how, and for whom**.
- Compare **market, command, and mixed systems**.
- Explain **opportunity cost** as the benefit of the next-best alternative forgone.

College Board CED: MKT-1.A · MKT-1.B

Resource

An input used to satisfy wants or produce goods and services.

College Board CED, MKT-1.A.2 · "established knowledge ... may not be scarce due to its non-rival nature"

Factors of production

Land, labor, and capital are common inputs used to produce output.

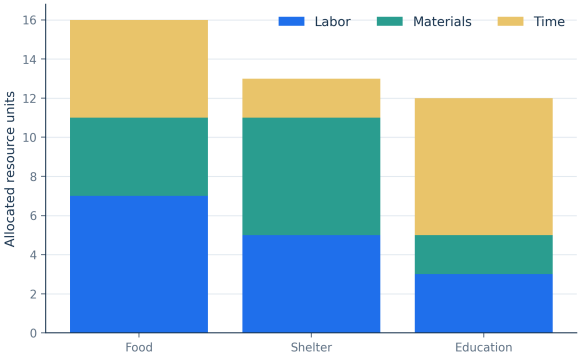
College Board CED, MKT-1.A.2 · "established knowledge ... may not be scarce due to its non-rival nature"

Established knowledge

It can often be used by several producers at once, so it may be non-rival.

College Board CED, MKT-1.A.2 · "established knowledge ... may not be scarce due to its non-rival nature"

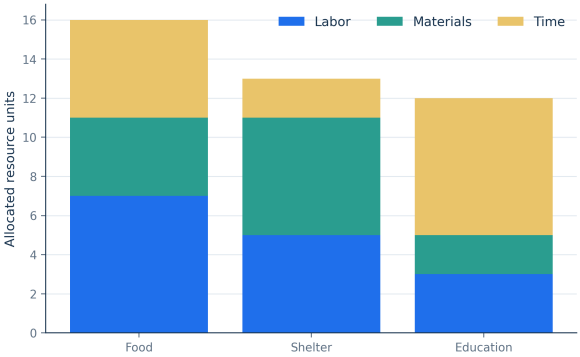
Scarcity turns resources into trade-offs



Look first.

What resources appear limited?

Scarcity turns resources into trade-offs

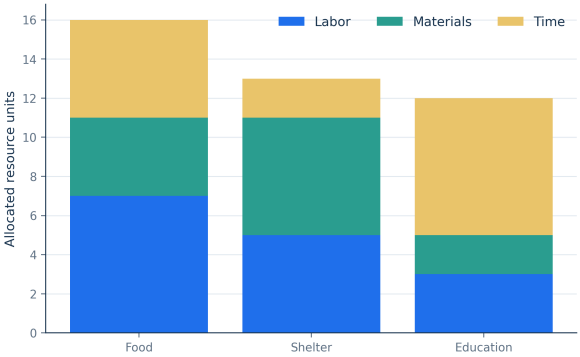


Three constraints

- labor
- materials
- time

Each input has competing uses.

Scarcity turns resources into trade-offs



Conclusion

Desired uses exceed available resources.Every decision creates an **economic trade-off**.

Scarcity

Resources are insufficient to satisfy all wants and needs.

College Board CED, MKT-1.A.1

Choice

A decision-maker selects one use for a scarce resource.

Opportunity cost

The benefit of the next-best alternative forgone.

CORE Econ, Unit 3.3 · MIT OpenCourseWare, Lecture 1

Every economic system answers three questions

What?

Which goods or
services will be made?

Every economic system answers three questions

What?

Which goods or services will be made?

How?

Which methods and inputs will be used?

Every economic system answers three questions

What?

Which goods or services will be made?

How?

Which methods and inputs will be used?

For whom?

Who will receive the output?

Every economic system answers three questions

What?

Which goods or services will be made?

How?

Which methods and inputs will be used?

For whom?

Who will receive the output?

Why all three matter

Fixed staff, time, and building capacity make all three questions unavoidable.

Economic systems coordinate scarce resources

A system is a way of coordinating choices.

Economic systems coordinate scarce resources

Market

Prices, incentives, and voluntary exchange coordinate decisions.

Economic systems coordinate scarce resources

Market

Prices, incentives, and voluntary exchange coordinate decisions.

Command

A central authority directs substantial resource use.

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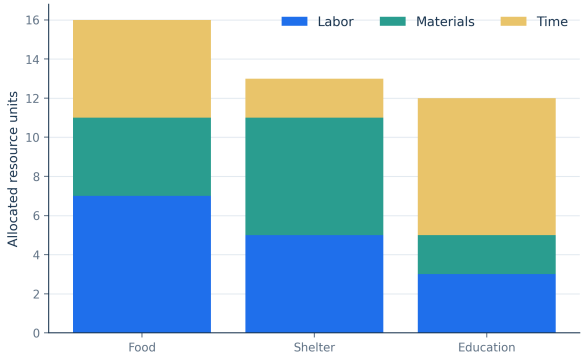
Mixed

Market and public institutions coordinate together.

Each system uses a coordinating mechanism to allocate scarce resources and distribute output.

College Board CED, MKT-1.B.2 · MIT OpenCourseWare, Lecture 1

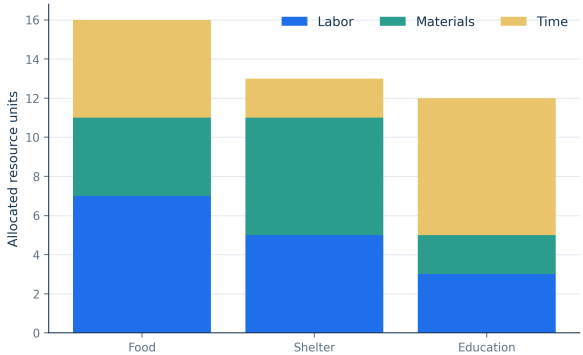
A resource bundle can support competing uses



Read the bars.

Which use receives the scarce bundle?

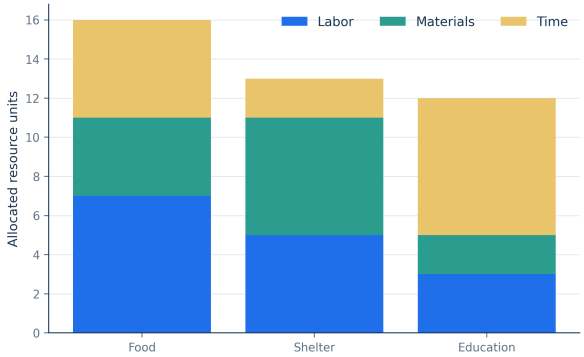
A resource bundle can support competing uses



Name the constraints.

- labor
- materials
- time

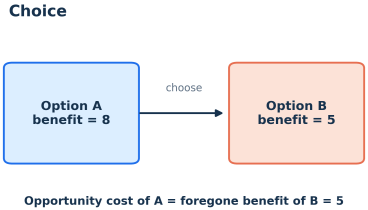
A resource bundle can support competing uses



Trade-off

More of one use requires less of another when the bundle is fixed.

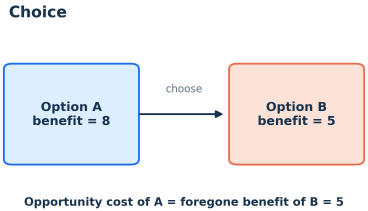
Opportunity cost names the next-best alternative



Choice

The decision-maker chooses A.

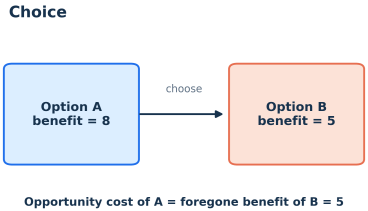
Opportunity cost names the next-best alternative



What is given up?

The next-best alternative is B.

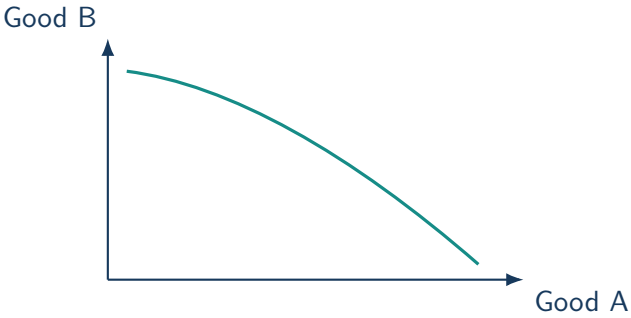
Opportunity cost names the next-best alternative



Opportunity cost

The benefit from B—the next-best alternative—is the opportunity cost of choosing A.

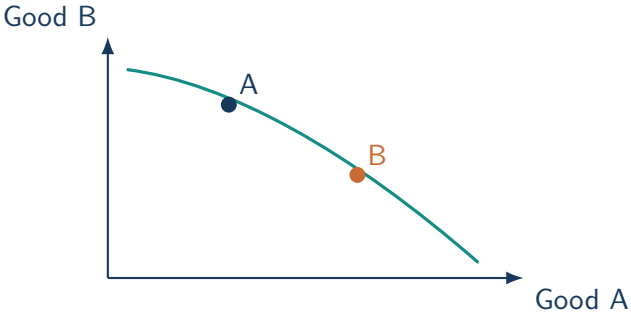
PPC preview: a movement along the boundary



Start with the boundary.

This PPC preview shows feasible combinations of two goods. Formal PPC analysis begins in Session 02.

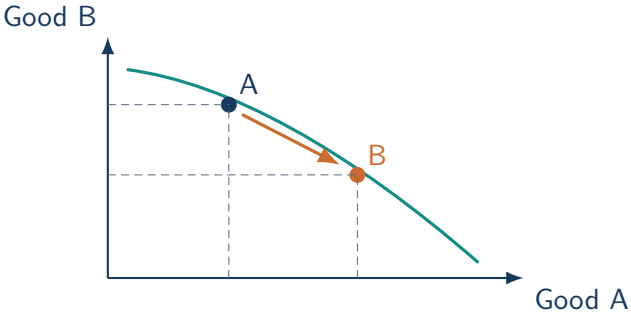
PPC preview: a movement along the boundary



Compare A and B.

Both points lie on the same boundary.

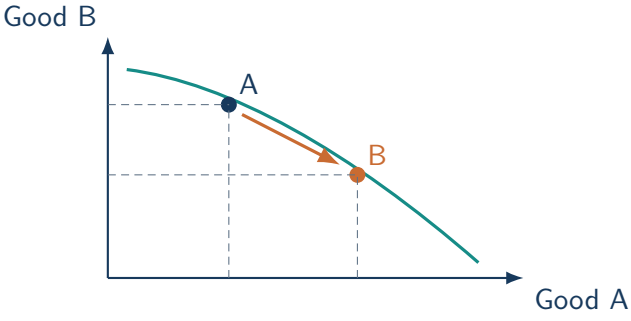
PPC preview: a movement along the boundary



Project to the axes.

Good A rises while Good B falls.

PPC preview: a movement along the boundary



Trade-off

Moving from A to B gives up Good B.

College Board CED, MKT-1.C.1 · Session 02 develops efficiency, growth, and numerical opportunity cost.

AP transfer: identify the scarce resource

One empty classroom must serve either a tutoring center or a rehearsal room. Which statement is most accurate?

1. Scarcity exists only if the school charges rent.
2. The room is scarce because it has alternative uses.
3. There is no scarcity if both activities are valuable.

Read the choices first.

AP transfer: identify the scarce resource

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Evidence

The classroom is fixed and has alternative uses.

AP transfer: identify the scarce resource

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2. The room is scarce because it has alternative uses.
3. There is no scarcity if both activities are valuable.

Answer: B

The room cannot serve both uses at once.

Exit ticket: state the causal chain

Because _____ is limited, choosing _____ means giving up _____.

Exit ticket: state the causal chain

Write one accurate causal sentence

Because _____ is limited, choosing _____ means giving up _____.

Use one scarce resource, one choice, and one foregone alternative.

Exit ticket: state the causal chain

Model answer

Because **classroom space** is limited, choosing **a tutoring center** means giving up **a rehearsal room**.

scarce resource $\longrightarrow$ **choice** $\longrightarrow$ **trade-off**